DECK 20 · WEEK 4 · AFTERNOON 5

Portfolio Management, MPT, CAPM & Risk

How the market prices risk — and how investors build portfolios to harvest the reward for bearing it

CFA Level I · Vol 9 Learning Modules 1–6 · FINAL DECK (20 of 20)

Roadmap — Portfolio Management, MPT, CAPM & Risk Management

6 sections · ~2 hours · Built bottom-up from CFA readings

01	**MPT foundations** risk aversion & utility · two-asset variance · correlation & diversification · efficient frontier · min-variance portfolios
02	**CAL, CML, CAPM & SML** capital allocation line · market portfolio · systematic vs idiosyncratic · β · CAPM · security market line · Sharpe / Treynor / M^2 / Jensen
03	**The portfolio management process** planning · execution · feedback · individual vs institutional investors · DB vs DC pensions · asset management industry
04	**The investment policy statement** IPS components · risk & return objectives · ability vs willingness · constraints (LTTLU) · Marie Gascon worked example
05	**Behavioral biases** cognitive errors (fixable) vs emotional biases (adapt) · belief perseverance · processing · loss aversion · disposition effect
06	**Risk management framework** risk governance · tolerance · budgeting · risk drivers vs exposures · closing the Deck 01 → Deck 20 arc

Learning objectives

By the end of this deck you should be able to:

1. Compute portfolio expected return and variance for a two-asset portfolio, and explain how correlation (not standard deviation alone) drives diversification benefits.

2. Derive an asset's expected return using the CAPM, distinguish systematic from idiosyncratic risk, and use the Security Market Line to identify under- and over-priced assets.

3. Compute and interpret the Sharpe ratio, Treynor ratio, M^2 and Jensen's alpha, and say which of them belongs on total risk (σ) and which on systematic risk (β).

4. Describe the three-step portfolio management process and the role of the IPS — including how to resolve the conflict between a client's ability and willingness to take risk.

5. Distinguish cognitive errors from emotional biases, and connect the risk management framework and the CAPM discount rate back to the DCF valuation of Deck 01.

SECTION 01

MPT foundations

Risk-averse investors require a higher expected return for every extra unit of variance they accept — utility encodes this trade-off numerically

Utility here is only an ordering device: a portfolio scoring 4 beats one scoring 2, but it is not twice as good, and scores cannot be compared across people.

$$U = E(R) - \tfrac{1}{2} \times A \times \sigma^2$$

where: U = utility; A = coefficient of risk aversion; σ^2 = variance (in decimals, not %). Higher A → steeper indifference curves → more averse.

1 **A > 0 · Risk-averse**
The assumed investor throughout. Demands higher return to accept variance. Indifference curves slope up and are convex — steeper as A rises. Curriculum examples run A = 2 (Kelly) to A = 4 (Jane).

2 **A = 0 · Risk-neutral**
Cares only about return, ignores risk. Indifference curves are horizontal. Picks the highest-return investment whatever its σ. A theoretical benchmark, not a client type.

3 **A < 0 · Risk-seeking**
Utility rises with variance, so indifference curves slope DOWN. Lottery-ticket and casino behaviour. Picks the highest risk AND highest return on offer.

Source: CFA Institute, Curriculum Vol 9, Learning Module 1, §4–5 "Risk Aversion" and "Utility Theory and Indifference Curves", pp. 12–17.

Portfolio return is a weighted average of asset returns — but portfolio variance depends on the correlation between assets, diversification's only free lunch

For a large N-asset portfolio, as N rises the w^2 variance terms vanish and average covariance dominates — the end-state of diversification.

$$E(R_p) = \Sigma\, w_i\, E(R_i) \qquad \sigma^2_p = \Sigma\, w_i^2\, \sigma_i^2 + \Sigma\, \Sigma_{i \neq j}\, w_i\, w_j\, Cov(R_i, R_j)$$

where: $Cov(R_i, R_j) = \rho_{ij} \times \sigma_i \times \sigma_j$. Return is linear in weights; risk is NOT — that asymmetry is what makes diversification work.

TWO-ASSET SPECIAL CASE

$\sigma^2_p = w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2\, w_1\, w_2\, \rho\, \sigma_1\, \sigma_2$

This is the workhorse formula you memorize for CFA L1 — every multi-asset result extends from this base.

LARGE-N LIMIT

As $N \to \infty$ with equal weights, $\sigma^2_p \to$ average pairwise covariance. Individual-asset variance ceases to matter — only the covariance structure does.

Source: CFA Institute, Curriculum Vol 9, Learning Module 1, §7 "Portfolio Risk & Portfolio of Two Risky Assets", pp. 21–23.

CFA LM 1 Example 3 — combining an 80% S&P 500 position with 20% Emerging Markets raises return and lowers risk at the same time

*Adding a *riskier* asset to a portfolio can REDUCE total portfolio risk when its correlation with the existing holdings is low enough — the classic MPT insight.*

Asset	Weight	E(R)	σ
S&P 500	80%	9.93%	16.21%
MSCI Emerging Mkts	20%	18.20%	33.11%
Covariance S&P / EM	–	–	0.0050
Portfolio (80/20)	100%	11.58%	15.10%

WHY IT WORKS

The covariance implies $\rho = 0.0050 / (0.1621 \times 0.3311) = 0.093$.

At near-zero correlation the cross term contributes only 0.00160 of the 0.02281 variance — so adding a 33% volatility asset still pulls total risk DOWN.

SOLUTION

Compute portfolio return (weighted avg) and variance using the two-asset formula.

1. $E(R_p) = 0.80 \times 9.93\% + 0.20 \times 18.20\% = 11.58\%$
2. $\sigma^2_p = (0.80^2 \times 0.1621^2) + (0.20^2 \times 0.3311^2) + (2 \times 0.80 \times 0.20 \times 0.0050)$
3. $= 0.01682 + 0.00439 + 0.00160 = 0.02281$
4. $\sigma_p = \sqrt{0.02281} = 15.10\%$

⇒ E(R)=11.58% and σ=15.10% — higher return AND lower risk than S&P alone.

Source: CFA Institute, Curriculum Vol 9, Learning Module 1, Example 3 "Return and Risk of a Two-Asset Portfolio", p. 23.

Correlation ρ is the lever: ρ = +1 erases diversification, ρ = −1 builds a risk-free portfolio, and every ρ below +1 delivers something in between

CFA LM 1 Example 4: two stocks with identical 10% return and 20% risk, combined 50/50. Only correlation changes — expected return stays 10% in every row.

$$\sigma^2\_p = w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2\, w_1\, w_2\, \rho\, \sigma_1\, \sigma_2$$

Correlation ρ	Portfolio variance	Portfolio σ_p	Diversification benefit
+1.00	0.04 (= weighted avg)	20.00%	None — σ equals weighted avg
+0.50	0.030	17.32%	2.68 pp risk reduction
0.00	0.020	14.14%	5.86 pp — no co-movement
−0.50	0.010	10.00%	Half the risk erased
−1.00	0.000	0.00%	Perfect hedge — risk-free portfolio

KEY RULE

For any ρ < +1, portfolio standard deviation is LESS than the weighted average of individual standard deviations. Equality only at ρ = +1.

Source: CFA Institute, Curriculum Vol 9, Learning Module 1, "Covariance and Correlation" and Example 4 "Effect of Correlation on Portfolio Risk", pp. 24–26.

The efficient frontier is the upper boundary of all portfolios with minimum variance at each expected return — every portfolio below it is dominated

Markowitz (1952): rational investors hold only portfolios on the efficient frontier. Everything inside the opportunity set, and the whole lower branch, is dominated.

THREE KEY PORTFOLIOS

Global Minimum-Variance Portfolio (GMVP): the single portfolio on the frontier with lowest possible risk.

Minimum-Variance Frontier: lowest σ for each level of E(R).

Efficient Frontier: the UPPER half of the minimum-variance frontier — from GMVP upward to the highest-return asset.

WHY THE LOWER HALF IS DOMINATED

Any portfolio on the lower branch has a mirror-image above it with the same σ but higher E(R).

No rational investor would ever choose the lower one — hence "dominated."

With risky assets only, the investor's optimum is where their highest indifference curve is TANGENT to this frontier. Add a risk-free asset and that tangency moves to the CAL.

ECONOMIC MEANING

Concave: doubling risk does NOT double return. Each extra unit of σ buys less E(R) than the one before it.

Source: CFA Institute, Curriculum Vol 9, Learning Module 1, §10 "Investment Opportunity Set & Minimum-Variance Portfolios", pp. 37–40.

SECTION 02

CAL, CML, CAPM & SML

Adding a risk-free asset produces the Capital Allocation Line — a straight line whose slope is the Sharpe ratio of the risky portfolio

The CAL dominates the efficient frontier of risky assets alone: for every point on the frontier there is a point on the CAL with the same σ and a higher return.

$$E(R_p) = R_f + [\ (E(R_i) - R_f) / \sigma_i\ ] \times \sigma_p$$

where: Intercept = R_f; slope = $(E(R_i) - R_f) / \sigma_i$ = Sharpe ratio = "market price of risk". Every portfolio on one CAL shares that slope.

AT R_f · LENDING

100% in the risk-free asset. $\sigma = 0$, return = R_f. Leftmost point on the CAL.

AT P · TANGENCY

100% in the risky portfolio. Point of tangency with the efficient frontier — the steepest CAL available, chosen WITHOUT reference to investor preferences.

BEYOND P · BORROWING

Borrow at R_f and invest >100% in P. Weight on the risk-free asset goes negative; return and σ both rise linearly.

WHERE THE INVESTOR ACTUALLY LANDS

Overlay the indifference curves: the optimal investor portfolio is where the highest attainable curve is TANGENT to the CAL.

Source: CFA Institute, Curriculum Vol 9, Learning Module 2, §2 "Portfolio of Risk-Free and Risky Assets", pp. 64–68.

When every investor holds the same optimal risky portfolio — the market portfolio — the CAL becomes the Capital Market Line, and two-fund separation obtains

Under homogeneous expectations every investor picks the same risky portfolio = the value-weighted market portfolio M. Differences in risk tolerance only change the R_f/M mix.

$$E(R_p) = R_f + [\,(E(R_m) - R_f) / \sigma_m\,] \times \sigma_p$$

where: Slope = Sharpe ratio of the market = equity risk premium ÷ market σ. This is the CAL of the market portfolio — the CML.

TWO-FUND SEPARATION THEOREM

Every rational investor holds only two funds:
 (1) the risk-free asset, and
 (2) the market portfolio M.

Risk tolerance dictates the split; asset selection is the SAME for everyone — the core insight behind index investing.

PASSIVE VS ACTIVE

If prices are informationally unbiased, replicating the index is optimal — that is the CML argument for passive investing.

Active managers must either beat the CML on skill or accept sitting below it after costs. The curriculum notes most analysts still believe active adds value; the debate is open.

[1] Axes matter: the CML plots total risk σ and holds only for efficient portfolios. The SML plots β and holds for any asset, efficient or not.

Source: CFA Institute, Curriculum Vol 9, Learning Module 2, §3 "The Capital Market Line", pp. 68–73; two-fund separation LM 1, p. 41.

Total risk = systematic + nonsystematic; only systematic risk is priced because diversification eliminates the rest for free

Note the equality is between VARIANCES, not standard deviations — the curriculum is explicit about that even though everyone says "total risk = systematic + nonsystematic".

$$\sigma^2_i = \beta^2_i \sigma^2_m + \sigma^2_e \qquad \text{(systematic + nonsystematic)}$$

Risk type	Curriculum synonyms	Examples given in the text	Diversifiable?
Systematic	market / non-diversifiable	Rates, inflation, cycles	No
Nonsystematic	company/industry-specific	A failed drug trial; a crash	Yes

WHY ONLY SYSTEMATIC RISK IS COMPENSATED

Suppose idiosyncratic risk paid a premium. Everyone would load up on it, diversify it away for free, and keep the premium. Demand would bid the price up until that premium reached zero. So in equilibrium it IS zero — and only non-diversifiable risk carries an expected-return premium.

Source: CFA Institute, Curriculum Vol 9, Learning Module 2, §5 "Systematic and Nonsystematic Risk" and "Pricing of Risk", pp. 77–79.

Beta measures how much an asset moves with the market — the only risk the market compensates you for

The curriculum's own warning: 12-month betas track today's risk but are noisy; 3–5-year betas are more accurate but may describe a company that no longer exists.

$$\beta_i = Cov(R_i, R_m) / \sigma^2_m = \rho_{i,m} \times \sigma_i / \sigma_m$$

where: β is both the slope coefficient in the market model $R_i = \alpha_i + \beta_i R_m + e_i$ and the standardised measure of systematic risk.

Beta	Interpretation	Curriculum example
$\beta = 0$	Uncorrelated with the market. E(R) = R_f in CAPM.	T-bill; gold (Ex. 5)
$0 < \beta < 1$	Less volatile than the market. Defensive.	Post-deal ANR, β 0.90
$\beta = 1$	Moves one-for-one with the market, by definition.	The market portfolio
$\beta > 1$	Amplifies the market. E(R) exceeds E(R_m).	Mueller Metals, β 1.41
$\beta < 0$	Moves against the market. E(R) below R_f.	Insurance; emerging mkt −0.24

PORTFOLIO BETA

$\beta_p = \Sigma w_i \times \beta_i$ — a weighted average, exactly like portfolio return. Risk-free asset β = 0, market β = 1.

Source: CFA Institute, Curriculum Vol 9, Learning Module 2, §7 "Calculation and Interpretation of Beta" and "Estimation of Beta", pp. 82–85.

The CAPM says expected return equals the risk-free rate plus beta times the equity risk premium — the single most-cited formula in modern finance

R_f is usually a government-bond yield; the ERP is the long-run excess return of equities over R_f — 4–6% in developed markets, and the single most argued-over input in finance.

$$E(R_i) = R_f + \beta_i \times [\, E(R_m) - R_f \,]$$

where: $E(R_m) - R_f$ = equity risk premium (ERP). The CAPM maps systematic risk (β) linearly into required return — and nothing else enters.

COST OF EQUITY FOR A CYCLICAL STOCK — THE SAME INPUTS WE USED IN DECK 09

Cyclical manufacturer, levered β = 1.20. Risk-free rate R_f = 4.0% on the 10-year government bond. Equity risk premium $E(R_m) - R_f$ = 6.0%.

1. Plug into CAPM: $E(R_i)$ = 4.0% + 1.20 × 6.0%
2. The beta premium is the STOCK'S premium, not the market's: 1.20 × 6.0% = 7.2%
3. Sum: 4.0% + 7.2% = 11.2%
4. Sensitivity: β = 1.50 → 13.0%; β = 0.80 → 8.8%. Every 0.10 of beta is worth 60bp here.

⇒ E(R) = 11.2% = required return = cost of equity = the r in the Deck 09 WACC and every DCF since Deck 01.

[1] SIX assumptions: (1) risk-averse, utility-maximising, rational investors; (2) frictionless markets — no costs, no taxes, borrow at R_f;
[2] (3) one single holding period; (4) homogeneous expectations; (5) infinitely divisible investments; (6) investors are price takers.

Source: CFA Institute, Curriculum Vol 9, Learning Module 2, §8 "Assumptions of the CAPM", pp. 87–88. Worked figures reuse the Deck 09 cost-of-capital inputs.

CFA LM2 Example 7 — Bajaj Auto has zero correlation with the market, so its CAPM expected return collapses back to the risk-free rate

Assumptions shared by both companies: R_f = 3%, $E(R_m)$ = 13%, σ_m = 23%. Only correlation with the market and σ_i differ between the two.

BAJAJ AUTO (INDIA) · ρ = 0.00

σ_i = 50%, $\rho_{i,m}$ = 0 with the market. Compute β and E(R).

1. $\beta = \rho \times \sigma_i / \sigma_m$
2. $= 0.00 \times 0.50 / 0.23 = 0.00$
3. $E(R) = 3\% + 0 \times (13\% - 3\%)$
4. $= 3.0\%$

⇒ E(R) = R_f = 3.0% — no systematic exposure = no premium.

MUELLER METALS (GERMANY) · ρ = 0.65

σ_i = 50%, $\rho_{i,m}$ = 0.65 with the market. Compute β and E(R).

1. $\beta = \rho \times \sigma_i / \sigma_m$
2. $= 0.65 \times 0.50 / 0.23 = 1.41$
3. $E(R) = 3\% + 1.41 \times (13\% - 3\%)$
4. $= 3\% + 14.1\% = 17.1\%$

⇒ E(R) = 17.1% — high correlation drives the premium.

TAKEAWAY

Identical total risk, σ = 50%. Only the CORRELATION differs — and it alone opens a 14.1pp gap in required return.

Source: CFA Institute, Curriculum Vol 9, Learning Module 2, Example 7 "Security Market Line and Expected Return", pp. 89–90.

The Security Market Line plots expected return against beta — assets above the line are underpriced, below the line are overpriced

Estimated return above the CAPM required return ⇒ $\alpha > 0$ ⇒ BUY. Below ⇒ $\alpha < 0$ ⇒ SELL. In equilibrium $\alpha = 0$ for every asset and every point sits exactly ON the line.

intercept = R_f · slope = E(R_m) − R_f · x-axis is BETA

ABOVE SML · UNDERPRICED

Estimated return > CAPM required return for the given β.

$\alpha > 0$.

Action: BUY — the market is paying you MORE than its equilibrium compensation for the risk taken.

ON THE SML · FAIRLY PRICED

Estimated return = CAPM required return for the given β.

$\alpha = 0$.

Action: HOLD at market weight. In CAPM equilibrium, every asset sits exactly on the SML — no free lunch.

BELOW SML · OVERPRICED

Estimated return < CAPM required return for the given β.

$\alpha < 0$.

Action: SELL / UNDERWEIGHT — compensation is inadequate for the β exposure. Long-only funds underweight; hedge funds short.

[1] The SML is not the CML. CML: total risk σ on the x-axis, efficient portfolios only. SML: systematic risk β on the x-axis, and it prices ANY asset, efficient or not.

Source: CFA Institute, Curriculum Vol 9, Learning Module 2, §8 "The Security Market Line", pp. 88–90.

Four appraisal measures, two denominators: Sharpe and M² divide by total risk, Treynor and Jensen's alpha divide by beta — and they can disagree

Which denominator is right depends on one question: is this portfolio the investor's entire wealth, or one sleeve of an already-diversified whole?

Measure	Formula	Denominator	Use it when
Sharpe ratio	$[E(R_p) - R_f] / \sigma_p$	Total risk σ	Portfolio is the investor's whole wealth
M^2 (Modigliani RAP)	$SR \times \sigma_m + R_f$	Total risk σ	Ranks like Sharpe; reads as a % return
Treynor ratio	$[E(R_p) - R_f] / \beta_p$	Systematic risk β	One sleeve of a well-diversified whole
Jensen's alpha	$R_p - [R_f + \beta_p \cdot ERP]$	Systematic risk β	Same as Treynor; also sizes the gap

Manager	Return	σ	β	Sharpe	Treynor	M² alpha	Jensen α
Manager X	10.0%	20.0%	1.10	0.35	0.064	0.65%	0.40%
Manager Y	11.0%	10.0%	0.70	0.80	0.114	9.20%	3.80%
Manager Z	12.0%	25.0%	0.60	0.36	0.150	0.84%	5.40%
Market M	9.0%	19.0%	1.00	0.32	0.060	0.00%	0.00%

THE DISAGREEMENT IS THE LESSON

Y wins on total risk, Z wins on beta risk. Hire Y to run everything; hire Z for one sleeve.

Source: CFA Institute, Curriculum Vol 9, Learning Module 2, §11 "Portfolio Performance Appraisal Measures" and Example 10, pp. 96–102.

SECTION 03

The portfolio process

Every professional portfolio is built in the same three steps — plan, execute, feed back — and the IPS anchors all three

The most-tested workflow in CFA L1 portfolio management. Sequence matters: everything downstream is defended by reference to the document written in step one.

1

STEP 1 · PLANNING

Understand the client's needs — objectives and constraints — through fact-finding and questionnaires, then write the IPS. It may name a benchmark, which step 3 later measures against. Review every three years or on any material change.

2

STEP 2 · EXECUTION

Asset allocation → security analysis → portfolio construction, then trading. Top-down sets the economic view, bottom-up picks the names. The curriculum is explicit: the asset allocation decision has the greatest impact on performance.

3

STEP 3 · FEEDBACK

Monitor and rebalance as weights drift from intended levels, then evaluate performance against the benchmark and report. The analysis may send you back to step one and rewrite the IPS — this is a loop, not a line.

Source: CFA Institute, Curriculum Vol 9, Learning Module 3, §4 "Steps in the Portfolio Management Process", pp. 133–136.

Liabilities, not preferences, drive institutional portfolios — which is why an endowment and a life insurer hold opposite assets in the same economy

Read the risk-tolerance column carefully — the curriculum rates DB pension plans as "typically quite high", which surprises most candidates. Long horizon, not caution, drives it.

Client type	Time horizon	Risk tolerance	Income needs	Liquidity needs
Individual investors	Varies	Varies	Varies	Varies
DB pension plans	Typically long	Quite high	High if mature	Varies by maturity
Endowments/foundations	Very long term	Typically high	Fund the spend	Quite low
Banks	Short term	Quite low	Pay depositors	High — withdrawals
Insurance companies	P&C short, life long	Quite low	Typically low	High — claims
Sovereign wealth funds	Varies by fund	Varies by fund	Varies by fund	Varies by fund

THE DIAGNOSTIC QUESTION

Ask what the investor OWES, and when — every column then follows. A bank owes deposits on demand: short horizon, low risk tolerance, high liquidity. An endowment owes a perpetual spending rate: very long horizon, high risk tolerance, low liquidity need. Individuals and sovereign funds read "varies" because their liabilities genuinely differ case by case.

[1] Yale endowment (June 2017): only 4.6% fixed income. MassMutual life insurer: about 80% in bonds, mortgages, loans and cash.

Source: CFA Institute, Curriculum Vol 9, Learning Module 3, §5 "Types of Investors", Exhibit 17, pp. 137–143. Portfolio weights from Exhibits 10 and 11, pp. 135–136.

DB and DC pensions place the investment risk on opposite sides of the ledger — the key distinction every portfolio manager must internalise

If the Deck 01 discount-rate framework ever felt abstract, a DB plan makes it concrete: the present value of the promised pensions IS a DCF, and funded ratio = assets ÷ that DCF.

DEFINED BENEFIT (DB)

The sponsor promises a specified benefit on retirement — the obligation is stated in retirement income owed to participants.

WHO BEARS THE RISK: the EMPLOYER, who is responsible for funding the benefits.

Managers must ensure assets are there when benefits fall due, often matching asset cash flows to expected pension payments with bonds. Horizon is indefinite if open to new members, finite if closed.

DEFINED CONTRIBUTION (DC)

The plan is in the employee's name and usually funded by both parties. A contribution is specified — a payout is not.

WHO BEARS THE RISK: the EMPLOYEE, who accepts BOTH the investment and the inflation risk.

US 401(k), UK group personal pensions, Australian superannuation. The employee is also responsible for there being enough in the plan at retirement.

THE GLOBAL SHIFT

Sponsors favour DC because it costs and risks the company less — moving both risks onto households.

[1] Global pension assets exceeded US$41 trillion at end-2017; the US, UK and Japan alone were more than 76% of that. Plan sponsors keep shifting from DB to DC.

Source: CFA Institute, Curriculum Vol 9, Learning Module 3, §5 "Individual Investors" and "Defined Benefit Pension Plans", pp. 137–139.

SECTION 04

The IPS

An IPS has eight sections plus appendices — but the two that carry the client's fingerprints are the objectives and the constraints

No single standard format exists. But in places it is a legal requirement: UK pension schemes need one under the Pensions Act 1995, s.35.

Section	What it contains
1. Introduction	Describes the client.
2. Statement of Purpose	States the purpose of the IPS itself.
3. Duties and Responsibilities	Of the client, the custodian of the assets, and the managers.
4. Procedures	How to keep the IPS current; how to respond to contingencies.
5. Investment Objectives	The objectives in investing — risk and return, which must agree.
6. Investment Constraints	Factors constraining those objectives — the "LTTLU" five.
7. Investment Guidelines	How policy is executed: leverage, derivatives, exclusions.
8. Evaluation and Review	How feedback on investment results is obtained.
Appendices A and B	Strategic asset allocation (policy portfolio); rebalancing.

Source: CFA Institute, Curriculum Vol 9, Learning Module 4, §2 "Major Components of an IPS", pp. 163–164.

Risk tolerance is the lower of ability (objective) and willingness (subjective) — never average them; always defer to the more conservative

The curriculum is direct: counsel the client, conclude at the LOWER of the two factors, and document the decision. Modifying personality is not the adviser's role.

Risk tolerance = MIN (Ability , Willingness)

	Willingness: BELOW avg	Willingness: ABOVE avg
Ability: ABOVE avg	Resolve — default to BELOW avg	CONSISTENT → above-average tolerance
Ability: BELOW avg	CONSISTENT → below-average tolerance	Resolve — default to BELOW avg

ABILITY = OBJECTIVE, WILLINGNESS = SUBJECTIVE

ABILITY drivers, mainly objective: time horizon, expected income, wealth relative to liabilities. A 20-year horizon beats a 2-year one; assets comfortably above liabilities beat a finely balanced position.

WILLINGNESS drivers, subjective: personality, self-esteem, independence of thought — gauged by discussion or a psychometric questionnaire such as the five-item Grable instrument in Exhibit 2.

Source: CFA Institute, Curriculum Vol 9, Learning Module 4, §3 "IPS Risk and Return Objectives", Exhibit 1, pp. 165–168.

The "RR LTTLU" template covers every IPS constraint — Liquidity, Time horizon, Tax, Legal/regulatory, Unique, on top of Risk & Return objectives

The mnemonic is a teaching device; the five constraints are the curriculum's own — liquidity, time horizon, tax concerns, legal and regulatory factors, unique circumstances.

Component	CFA definition	Illustration — Sophie, 45, UK DC
RETURN objective	Absolute or relative, nominal or real	Real 4.5% p.a. over 20 years
RISK objective	Absolute (σ, VaR) or relative (tracking)	Max 15% σ; 25% drawdown limit
LIQUIDITY	What must be withdrawn from the portfolio	£20k emergency fund in cash
TIME horizon	Accumulation period, or until things change	Two-stage: 20y, then drawdown
TAXES	Taxed or not; income versus capital gains	UK higher rate; pension relief
LEGAL / regulatory	Portfolio limits, self-investment caps	UK DC rules; lump sum from 55
UNIQUE circumstances	Beliefs, values, ESG, concentrated stakes	Employer stock capped at 10%

THE CURRICULUM NAMES SIX ESG APPROACHES

Negative screening (exclude) · positive screening (best-in-class) · ESG integration (systematic, in allocation and selection) · thematic investing · engagement and active ownership · impact investing.

Source: Definitions: CFA Institute, Curriculum Vol 9, Learning Module 4, §4 "IPS Constraints", pp. 172–176. Sophie is an instructor-built illustration, not a curriculum case.

CFA LM4 Example 4 — Marie Gascon's case solves the annual return she needs to fund a €2m retirement in 20 years at 2% inflation

The return objective falls out of the constraints — target capital, horizon, inflation — not out of a market view. This is Deck 01 time-value arithmetic, used backwards.

MARIE GASCON · RETIRING AT 50 · 20-YEAR HORIZON

Savings €1,000,000. €100,000 is set aside as an emergency fund in cash, so €900,000 is invested for retirement. She estimates €2,000,000 in today's money will fund her retirement income needs. Inflation is expected to average 2% p.a. French pension contributions and returns are tax-exempt; distributions at retirement are taxable.

1. Inflate the target: €2,000,000 × 1.02^20 = €2,971,895 in money of the day.
2. Required nominal CAGR on €900,000: (€2,971,895 / €900,000)^(1/20) − 1.
3. Ratio = 3.3021; 3.3021^(1/20) = 1.0615 → 6.15%, which rounds to the curriculum's 6.2%.
4. Cross-check in real terms: (€2,000,000 / €900,000)^(1/20) − 1 = 4.07% real p.a.
5. Verify at the curriculum's 6.2%: €900,000 × 1.062^20 = €2,997,318, just above target. ✓

⇒ Required return ≈ 6.2% nominal (≈ 4.1% real) — now build a portfolio that plausibly delivers it inside Marie's risk tolerance.

Source: CFA Institute, Curriculum Vol 9, Learning Module 4, Example 4 "The Case of Marie Gascon: Return Objectives", p. 171.

SECTION 05

Behavioral biases

Behavioral biases split cleanly into cognitive errors — fixable with better information — and emotional biases, which you can only recognise and adapt around

The distinction drives the remedy: cognitive errors respond to better information, education and advice. Emotional biases arise spontaneously and often can only be recognised and adapted to.

COGNITIVE ERRORS

SOURCE: faulty reasoning. Mistakes made while processing information.

SPLIT: belief-perseverance (conservatism, confirmation, representativeness, illusion of control, hindsight) and processing (anchoring, mental accounting, framing, availability).

FIX: yes — with better information, education, advice, systematic decision frameworks.

EMOTIONAL BIASES

SOURCE: impulses, intuitions, feelings. Not conscious errors — they arise spontaneously.

SIX: loss aversion, overconfidence, self-control, status quo, endowment, regret aversion.

FIX: no — recognize and ADAPT. Structural portfolio rules (rebalancing thresholds, sell-discipline, DCA) that remove discretion from the client.

RULE OF THUMB

Cognitive → educate. Emotional → automate. What you cannot teach away, engineer out of the decision path.

Source: CFA Institute, Curriculum Vol 9, Learning Module 5, §2–3 "Behavioral Bias Categories" and "Cognitive Errors", pp. 206–207.

Fifteen named biases in three groups — five belief-perseverance, four processing, six emotional — and the group tells you how to treat each one

Count them: 5 belief-perseverance + 4 processing = 9 cognitive, plus 6 emotional = 15. The first two groups are both cognitive — it is not a 2×2, it is a split of a split.

BELIEF PERSEVERANCE

Cognitive · five biases

Conservatism — underweight new information vs priors.

Confirmation — seek what confirms; ignore what contradicts.

Representativeness — judge on similarity; neglect base rates.

Illusion of control — overestimate your own influence.

Hindsight — "I knew it all along", after the fact.

PROCESSING ERRORS

Cognitive · four biases

Anchoring and adjustment — estimates stay too close to the first number seen.

Mental accounting — treat fungible money as separate pots.

Framing — answer changes with the presentation.

Availability — overweight recent or memorable events.

EMOTIONAL BIASES

Six biases · adapt, do not educate

Loss aversion — hold losers, sell winners.

Overconfidence — intervals too narrow.

Self-control — short-term gratification wins.

Status quo — leave the portfolio alone.

Endowment — value what you own too highly.

Regret aversion — avoid decisions you may regret.

Source: CFA Institute, Curriculum Vol 9, Learning Module 5, §3 "Cognitive Errors" and §4 "Emotional Biases", pp. 206–228.

Loss aversion inverts the textbook investor: risk-averse in gains, but risk-SEEKING in losses — which is why people double down on what is already falling

Kahneman & Tversky (1979): the value function is S-shaped and asymmetric around a reference point — a loss weighs more than a gain of the same absolute size.

THE S-SHAPED VALUE FUNCTION

Above the reference point (gains): CONCAVE → risk-avoiding. The first €10 of gain is worth more than the second €10.

Below it (losses): CONVEX → risk-SEEKING. The investor will accept more risk to avoid realising a loss than to make a gain.

Asymmetry: for the same absolute move, the loss side has the greater impact on utility. The curriculum states this qualitatively and puts no number on it.

THE DISPOSITION EFFECT

The name for the behaviour: hold losers too long, hoping to break even, and sell winners too quickly, in case the gain evaporates.

Consequence per the curriculum: the resulting portfolio may be RISKIER than the one the investor's own risk–return objectives call for.

Remedy: a disciplined process. Analyse the position and weigh the actual probabilities of future gains and losses — not the entry price.

PM IMPLICATION

Automate the sell discipline. Discretion at the exit is where loss aversion destroys most value.

Source: CFA Institute, Curriculum Vol 9, Learning Module 5, §4 "Loss-Aversion Bias" and Example 10, pp. 219–221.

SECTION 06

Risk management

Risk management is not risk minimisation — it is choosing and sizing the risks that fit the enterprise's tolerance, then monitoring them continuously

"The Doctrine of No Surprises" does not mean predicting the event. It means that when the event happens, its impact on the portfolio has already been quantified and discussed.

Framework element	What it does
1. Risk governance	Board-level and top-down: sets goals, risk tolerance, oversight.
2. Risk identification & measurement	Qualitative and quantitative: risk drivers, exposures, metrics.
3. Risk infrastructure	People and systems: data, models, a scenario engine, reporting.
4. Policies and processes	Limits and procedures that push governance into daily decisions.
5. Monitoring, mitigation, management	Continuous. Spot exposure drifting from tolerance, then act on it.
6. Communications	Continual, at every level, with a feedback loop to the board.
7. Strategic analysis and integration	Risk management as an offensive tool — better decisions, more value.

THREE THINGS EVERYONE CALLS "RISK"

Risk DRIVER = the underlying uncertainty. Risk POSITION = the risky action taken. Risk EXPOSURE = the valuation change that may result. In the curriculum's yen example those are ±1%, ¥1,000,000, and ±¥10,000 — and exposure = position × driver.

Source: CFA Institute, Curriculum Vol 9, Learning Module 6, §2–3 "Risk Management Process" and "Risk Management Framework", pp. 243–247.

Risk tolerance says which risks are acceptable; risk budgeting says how much of that tolerance each strategy may spend — appetite becomes positions

Tolerance should be chosen and communicated BEFORE a crisis. The curriculum notes many organisations only do it afterwards — better than never, but much like insuring after the fire.

RISK TOLERANCE

WHICH risks are acceptable, and at what level?

A board decision, built from two analyses:
- · Inside view — what shortfalls would make us fail
- · Outside view — which risk drivers are we exposed to

RISK BUDGETING

HOW MUCH of that tolerance does each strategy, asset or factor get to spend?

Four single-dimension metrics named in the text: standard deviation, beta, VaR, scenario loss.

It forces every risky decision to compete on return per unit of risk.

1ST LINE · BUSINESS

PMs, traders, front office. Own the risks; make the decisions.

2ND LINE · RISK

Risk team, CRO, compliance. Independent oversight of the first line.

3RD LINE · AUDIT

Internal audit. Periodic review of both lines; reports to the board.

[1] The three-lines-of-defence row above is standard bank and asset-manager practice. It is NOT part of the CFA Level I curriculum.

Source: CFA Institute, Curriculum Vol 9, Learning Module 6, §5–6 "Risk Tolerance" and "Risk Budgeting", pp. 254–258.

The full arc from Deck 01 to Deck 20 collapses into one statement: the discount rate we assumed on day one is the equilibrium price of systematic risk

The "r" in every DCF since Deck 01 stopped being an assumption twenty minutes ago. MPT and the CAPM tell you exactly what it must be in equilibrium, and why.

$$PV = \Sigma\ CF_t\ /\ (1+r)^t \quad \text{with} \quad r = R\_f + \beta \times [E(R\_m) - R\_f]$$

where: Deck 01 on the left, Deck 20 on the right. The two bookends of the course are the same equation seen from opposite sides.

Decks	What they covered	What they contributed to the DCF
01	TVM and DCF foundations	Defined PV, NPV, IRR — and took r as given
02–05	Quant methods and economics	Distributions, regression, rates, inflation, FX
06–08	Financial statement analysis	Supplied and stress-tested the cash flows CF_t
09–12	Corporate issuers and equity	WACC and the cost of equity; forecast the payouts
13–16	Fixed income	Priced the risk-free leg and the credit spread
17–19	Derivatives and alternatives	Hedged, replicated, and priced illiquidity
20	MPT and CAPM (this deck)	Derived r = R_f + β·ERP — the last missing piece

Source: Synthesis across the 20-deck course — see ../curriculum_map.md. Ties back to Deck 01 slides 5–8 and Deck 09 slide 30.

KEY TAKEAWAYS

Deck 20 — Portfolio Management, MPT, CAPM & Risk Management

- Return adds linearly across a portfolio; risk does not. For any $\rho < +1$, portfolio σ is below the weighted average — that gap is the whole of diversification.
- Efficient frontier = the min-variance frontier ABOVE the GMVP. Add a risk-free asset and you get the CAL; make the risky portfolio the market and you get the CML.
- Only systematic risk is priced. $E(R) = R_f + \beta \times ERP$. Three lines, three x-axes: CAL and CML on σ, SML on β. Above the SML = underpriced, below = overpriced.
- Sharpe and M^2 use total risk σ; Treynor and Jensen's alpha use β. Use σ when the portfolio is all the investor has; use β when it is one sleeve of many.
- Process = Plan, Execute, Feed back, and back to Plan. The IPS has eight sections plus appendices, and separates objectives from the five LTTLU constraints.
- Risk tolerance = the LOWER of ability and willingness, never the average. Cognitive biases you educate away; emotional ones you design the portfolio around.
- Risk management is not minimisation. Seven framework elements; tolerance is the board's sentence, the budget is its arithmetic in σ, β, VaR or scenario loss.
- And the arc closes: the r you assumed in Deck 01 is $R_f + \beta \cdot ERP$. Twenty decks, one equation. Everything else was learning what to put into it. Thank you.